

Year 7 · Chapter 8 · Area and volume: answers (teacher copy)

8.01 Units of measurement (booklet p. 3)

WE DO examples

1. Ex 1: 3.2 km
2. Ex 2: 42 g
3. Ex 3: 8300 mL
4. Ex 4: 4 h
5. Ex 5: 54 000 s
6. Ex 6: 6 307 200 min (B)

Set A

1. 3.2 km
2. 9040 mm
3. 7.15 m
4. 42 g
5. 0.0035 t
6. 0.125 kL

Set B

1. 8300 mL
2. 4 200 000 mL
3. 4 h
4. 900 min
5. 60 h
6. 17 kg

Set C

1. 54 000 s
2. 65 000 cm
3. 4.5 kg
4. 200 min
5. 0.75 L
6. 10 080 min

Set D

1. 6 307 200 min
2. 4.2 kg
3. 7.2 L
4. 950 m, 1 100 000 mm, 1.2 km, 125 000 cm

Extension

1. about 36 792 000
2. 11.6 days

Exit ticket

1. Level 1: 450 cm
2. Level 2: 2.75 kg
3. Level 3: 210 min

8.02 Perimeter (booklet p. 7)

WE DO examples

1. Ex 1: 52 m
2. Ex 2: 36 mm
3. Ex 3: 78 cm
4. Ex 4: 650 cm
5. Ex 5: 128 mm
6. Ex 6: 11.5 cm

Set A

1. 46 cm
2. 68 m
3. 36 mm
4. 38 cm
5. 13.6 m
6. 30 cm

Set B

1. 78 cm
2. 52 m
3. 128 mm

Set C

1. 136 m
2. 74 cm
3. 190 mm
4. 650 cm
5. 11.5 cm
6. 6 m

Set D

1. \$2597.00
2. 9.5 cm
3. 5 (1×9, 2×8, 3×7, 4×6, 5×5)
4. 66 cm

Extension

1. 10 cm by 15 cm
2. 14 cm

Exit ticket

1. Level 1: 38 cm
2. Level 2: 28.5 m
3. Level 3: 15 mm

8.03 Circumference of a circle (booklet p. 11)

WE DO examples

1. Ex 1: 6.3 cm
2. Ex 2: 62.8 mm
3. Ex 3: 44.0 cm
4. Ex 4: 175.9 mm
5. Ex 5: 207.3 m
6. Ex 6: 15.9 cm

Set A

1. 10 cm
2. 24 m
3. 7 mm
4. 1.6 km
5. 42 cm
6. 18 m

Set B

1. 6.3 cm
2. 44.0 cm
3. 22.0 m
4. 94.2 mm
5. 4.7 km
6. 314.2 cm

Set C

1. 62.8 mm
2. 175.9 mm
3. 18.8 cm
4. 28.3 m
5. 78.5 cm
6. 3.1 km

Set D

1. 207.3 m
2. 15.9 cm
3. 30.8 cm
4. 12 (circumference 377.0 cm)

Extension

1. 36.0 m
2. 401.1 m

Exit ticket

1. Level 1: 16 cm
2. Level 2: 28.3 m
3. Level 3: 22.0 cm

8.04 Area units (booklet p. 15)**WE DO examples**

1. Ex 1: a 1 cm square is 10 mm by 10 mm
2. Ex 2: 91 000
3. Ex 3: 250
4. Ex 4: 17.5
5. Ex 5: 0.24
6. Ex 6: 8 600 000

Set A

1. 300 mm²
2. 250 mm²
3. 91 000 cm²
4. 40 000 cm²
5. 2 000 000 m²
6. 50 000 m²

Set B

1. 2.4 cm²
2. 17.5 m²
3. 5 ha
4. 8.5 cm²
5. 3.5 km²
6. 0.64 m²

Set C

1. 0.24 m²
2. 8 600 000 mm²
3. 7 ha
4. 125 000 m²
5. 3500 cm²
6. 0.0072 m²

Set D

1. mm² or cm²; m²; ha; km²
2. 3.2 m² (30 000 cm² = 3 m²)
3. 150 tiles
4. 240 ha

Extension

1. 1 000 000 mm²
2. 0.71 ha

Exit ticket

1. Level 1: 500 mm²
2. Level 2: 3 m²
3. Level 3: 1 500 000 mm²

8.05 Area of squares and rectangles

(booklet p. 19)

WE DO examples

1. Ex 1: 16 m²
2. Ex 2: 24 cm²
3. Ex 3: 162 m²
4. Ex 4: 1.5 m²
5. Ex 5: 9 cm
6. Ex 6: 6 m

Set A

1. 16 m²
2. 24 cm²
3. 162 m²
4. 60 mm²
5. 49 cm²
6. 3 m²

Set B

1. 120 cm²
2. 121 m²
3. 7 m²
4. 0.25 km²
5. 1000 mm²
6. 0.6 m²

Set C

1. 9 cm
2. 6 m
3. 3 cm
4. 1.2 m
5. 1.5 m²
6. 81 cm²

Set D

1. \$936
2. 108 tiles
3. 8 cm by 5 cm
4. it is 4 times as large

Extension

1. 24 m²
2. 6 cm by 6 cm (36 cm²)

Exit ticket

1. Level 1: 54 cm²
2. Level 2: 6.25 m²
3. Level 3: 12 m

8.06 Area of a triangle (booklet p. 23)**WE DO examples**

1. Ex 1: 360 mm²
2. Ex 2: 112 mm²
3. Ex 3: 5.2 cm²
4. Ex 4: 187 km²
5. Ex 5: 8 cm
6. Ex 6: two copies of the triangle make a rectangle (or parallelogram)

Set A

1. 360 mm²
2. 30 cm²
3. 20 m²
4. 42 cm²
5. 5.2 cm²
6. 150 m²

Set B

1. 12 cm^2
2. 112 mm^2
3. 54 m^2
4. 17.5 cm^2
5. 1.8 m^2
6. 2500 cm^2

Set C

1. 187 km^2
2. 40 m^2
3. 54 cm^2
4. 8 cm
5. 5 m
6. 30 cm^2

Set D

1. 8 m^2
2. 4 bags
3. 92 cm^2
4. the second is twice the first (20 cm^2 vs 40 cm^2)

Extension

1. 27 m^2
2. 24 square units

Exit ticket

1. Level 1: 35 m^2
2. Level 2: 8 cm^2
3. Level 3: 8 cm

8.07 Area of a parallelogram (booklet p. 27)**WE DO examples**

1. Ex 1: 120 m^2
2. Ex 2: 828 cm^2
3. Ex 3: 68 mm^2
4. Ex 4: cutting off a triangle and moving it makes a rectangle
5. Ex 5: 6 cm
6. Ex 6: 50 cm^2

Set A

1. 120 m^2
2. 828 cm^2
3. 68 mm^2
4. 60 cm^2
5. 180 m^2
6. 31.5 cm^2

Set B

1. 60 cm^2
2. 9 m^2
3. 300 mm^2
4. 28 km^2
5. 0.96 m^2
6. 450 cm^2

Set C

1. 6 cm
2. 9 m
3. 2 m
4. 50 cm^2
5. 98 mm^2
6. parallelogram 48 cm^2 , triangle 24 cm^2

Set D

1. \$882
2. e.g. 6×4 and 8×3
3. 80 cm^2
4. 9 cm

Extension

1. 32 m^2
2. 12 cm

Exit ticket

1. Level 1: 36 cm^2
2. Level 2: 4 m^2
3. Level 3: 12.5 m

8.08 Area of composite shapes (booklet p. 31)**WE DO examples**

1. Ex 1: 23 cm^2
2. Ex 2: 71 m^2
3. Ex 3: 30 m^2
4. Ex 4: 360 m^2
5. Ex 5: 6 L
6. Ex 6: 37.5 mm^2

Set A

1. 71 m^2
2. 72 cm^2
3. 105 m^2

Set B

1. 23 cm^2
2. 30 m^2
3. 78 m^2

Set C

1. 360 m^2
2. 37.5 mm^2
3. 272.4 mm^2
4. 84 cm^2
5. 80 m^2
6. 55 cm^2

Set D

1. 6 L
2. 630 cm^2
3. 21 m^2
4. \$798

Extension

1. 57 m^2
2. 45 cm^2

Exit ticket

1. Level 1: 46 cm^2
2. Level 2: 22 m^2
3. Level 3: 3 L

8.09 Prisms, cross-sections and views

(booklet p. 35)

WE DO examples

1. Ex 1: yes; rectangle
2. Ex 2: yes; triangle
3. Ex 3: L-shape
4. Ex 4: no: the cross-section changes and the ends are not the same
5. Ex 5: front: L-shape; left: rectangle; top: rectangle
6. Ex 6: right: rectangle; front: pentagon (house shape); top: rectangle with a line

Set A

1. yes, rectangle
2. yes, triangle
3. yes, trapezium
4. yes, hexagon
5. no
6. not a prism (curved), but it has a circular cross-section

Set B

1. L-shape
2. pentagon
3. T-shape

Set C

1. L-shape; rectangle; rectangle
2. rectangle; pentagon; rectangle split by a line

Set D

1. T-shape; rectangle; rectangle
2. a cylinder
3. a cube (or sphere)
4. a rectangle

Extension

1. a square pyramid
2. three squares in a row of heights 1, 2, 3 (or one tall column, depending on view)

Exit ticket

1. Level 1: No: it has curved sides (it is like a prism with a circle cross-section)
2. Level 2: hexagon
3. Level 3: a rectangle with a line along the middle

8.10 Volume units (booklet p. 39)

WE DO examples

1. Ex 1: a 1 cm cube is $10 \times 10 \times 10$ mm
2. Ex 2: 20 000
3. Ex 3: 500 000
4. Ex 4: 7.5
5. Ex 5: 0.00023
6. Ex 6: 60 cubes

Set A

1. 20 000 mm³
2. 3000 mm³
3. 2 000 000 cm³
4. 500 000 cm³
5. 4500 mm³
6. 1 200 000 cm³

Set B

1. 7.5 cm³
2. 4 cm³
3. 3 m³
4. 0.25 m³
5. 0.6 cm³
6. 0.08 m³

Set C

1. 0.00023 m³
2. 4 000 000 mm³
3. 5 000 000 000 m³
4. 750 000 cm³
5. 0.0125 m³
6. 0.000009 m³

Set D

1. 60
2. 30
3. 15 000 cm³, 0.02 m³, 25 000 000 mm³
4. cm³; m³; cm³

Extension

1. 350 000 mm³
2. 1000

Exit ticket

1. Level 1: 6000 mm³
2. Level 2: 2.5 m³
3. Level 3: 1.5 cm³

8.11 Volume of a rectangular prism (booklet

p. 43)

WE DO examples

1. Ex 1: 80 m³
2. Ex 2: 320 cm³
3. Ex 3: 672 mm³
4. Ex 4: 3000 cm³
5. Ex 5: 4 cm
6. Ex 6: 5 cm

Set A

1. 80 m³
2. 320 cm³
3. 672 mm³
4. 360 m³
5. 3000 cm³
6. 27 cm³

Set B

1. 60 cm³
2. 125 m³
3. 5 m³
4. 300 mm³
5. 120 cm³
6. 0.3 m³

Set C

1. 4 cm
2. 5 cm
3. 6 m
4. 10 mm
5. 0.03 m³
6. 8 times

Set D

1. 74.88 m³
2. 72 cubes
3. 0.96 m³
4. e.g. $1 \times 1 \times 48$, $2 \times 4 \times 6$, $3 \times 4 \times 4$

Extension

1. 540 cm³
2. 108 s

Exit ticket

1. Level 1: 60 cm^3
2. Level 2: 216 m^3
3. Level 3: 6 cm

8.12 Volume of a prism (booklet p. 47)

WE DO examples

1. Ex 1: 554.472 cm^3
2. Ex 2: 2200 mm^3
3. Ex 3: 570 m^3
4. Ex 4: 280 cm^3
5. Ex 5: 120 cm^3
6. Ex 6: 30 m^2

Set A

1. 100 cm^3
2. 280 cm^3
3. 36 m^3
4. 30 mm^3
5. 50 cm^3
6. 480 m^3

Set B

1. 120 cm^3
2. 2200 mm^3
3. 144 m^3

Set C

1. 554.472 cm^3
2. 570 m^3
3. 540 cm^3

Set D

1. 5.4 m^3
2. 0.6 m^3
3. 30 m^2
4. $300\,000 \text{ cm}^3$

Extension

1. 1600 m^3
2. $375\,000 \text{ L}$

Exit ticket

1. Level 1: 90 cm^3
2. Level 2: 90 m^3
3. Level 3: 8 cm

8.13 Capacity (booklet p. 51)

WE DO examples

1. Ex 1: 6 L
2. Ex 2: 15 cm (D)
3. Ex 3: 3500 L
4. Ex 4: $80\,000 \text{ L}$
5. Ex 5: $2\,000\,000\,000 \text{ ML}$
6. Ex 6: 8 cups

Set A

1. 500 mL
2. 2 L
3. 3500 L
4. 200 L
5. 4.5 kL
6. $2\,000\,000 \text{ L}$

Set B

1. 6 L
2. $80\,000 \text{ L}$
3. 0.32 L
4. 60 L
5. 1000 L
6. 1 L

Set C

1. $0.672 \text{ mL} \rightarrow 0.000672 \text{ L}$
2. $360\,000 \text{ L}$
3. 3 L
4. 0.0022 L
5. $570\,000 \text{ L}$
6. $15\,625 \text{ L}$

Set D

1. 15 cm
2. $2\,000\,000\,000 \text{ ML}$
3. 63 L
4. 10 cm

Extension

1. 50 mm
2. $1\frac{1}{3} \text{ days}$

Exit ticket

1. Level 1: 2.5 L
2. Level 2: 5 L
3. Level 3: 600 L